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REVIEW

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# Research progress of spectroscopic detection technology in agricultural product quality detection

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**Abstract**

With the continuous improvement of smart agriculture and food safety requirements, accurate detection of agricultural product quality has become a key link in the entire agricultural industry chain. As an important non-destructive detection method for agricultural product quality, spectroscopic detection technology has achieved significant progress in detection accuracy, speed, and application scope in recent years. This paper systematically reviews the application research of technologies such as hyperspectral imaging (HSI), visible-near-infrared spectroscopy (Vis-NIR), near-infrared spectroscopy (NIRS), mid-infrared spectroscopy (MIRS), Raman spectroscopy (Raman), X-ray computed tomography (X-ray CT), surface-enhanced Raman spectroscopy (SERS), fluorescence imaging, terahertz time-domain spectroscopy (THz-TDS), and electrochemical impedance spectroscopy (EIS) in the quality detection of various agricultural products. It focuses on analyzing the latest achievements of these technologies in external quality evaluation, internal quality evaluation, nutrient component analysis, freshness and safety detection, and adulteration identification of agricultural products. This review also summarizes the challenges faced by existing spectroscopic detection technologies and proposes prospects for future development directions, including the development of miniaturized equipment, multi-technology integration, and intelligent analysis, aiming to provide comprehensive references for further research and application of spectroscopic technology in the field of agricultural product quality detection.

**Keywords** Spectroscopic technology, Agricultural product detection, Hyperspectral imaging (HSI), Quality evaluation

## Introduction

In the current market, the quality of agricultural products is directly related to consumers' health and the economic benefits of agricultural production. Quality assessment is an important task in the agricultural product industry chain and plays a crucial role in improving its economic value [1]. With the global population growth and the continuous improvement of living standards, the market demand for high-quality, green, and safe agricultural products has increased significantly, which has rapidly promoted the development of the field of agricultural product quality detection technology. Traditional quality detection methods usually require complex sample pretreatment, are time-consuming, and cause environmental pollution, making it difficult to meet the needs of the modern agricultural product industry for rapid and non-destructive detection. In recent years, with the rapid development of optical technology, sensor technology, and computer technology [2], spectroscopic detection technology has shown great potential in agricultural product quality detection due to its advantages of rapidity, non-destructiveness, and high accuracy [3, 4], and as shown in Fig. 1.

As an advanced detection method based on the principle of interaction between substances and

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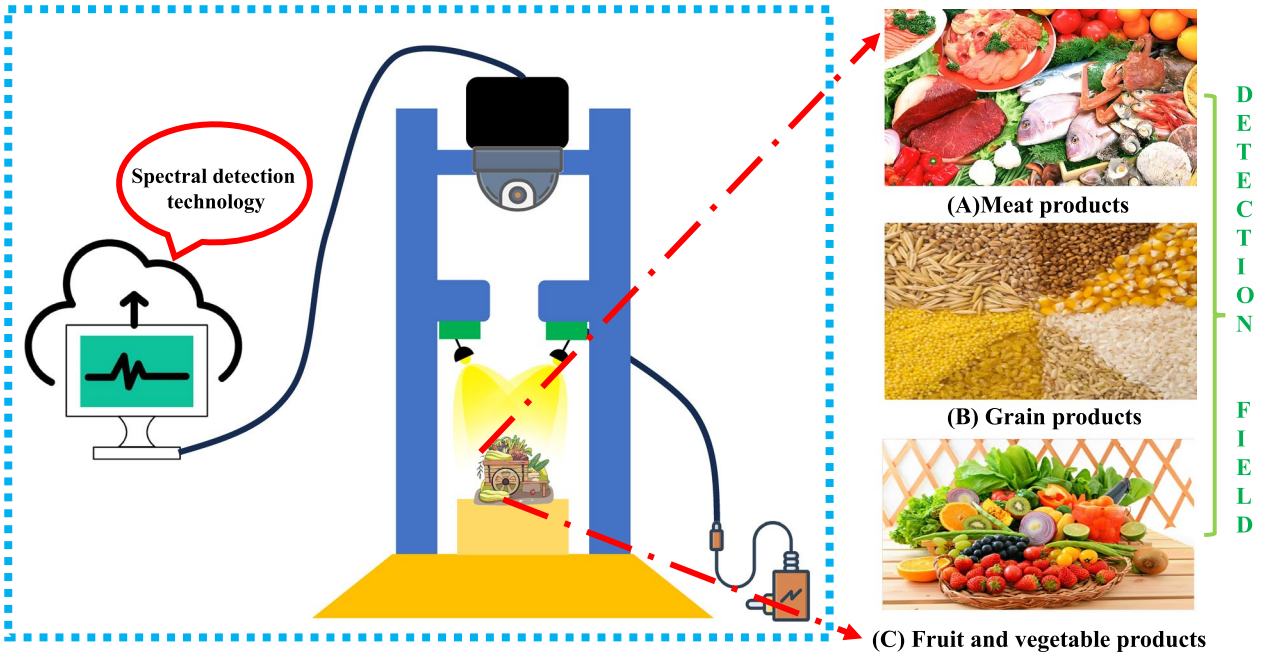



**Fig. 1** Application of spectral detection technology

electromagnetic waves, spectroscopic detection technology has attracted widespread attention in the field of agricultural product quality detection in recent years due to its advantages of non-contact, rapid response, and high sensitivity [5, 6]. By collecting spectral information on the surface and inside of agricultural products, this technology can effectively realize the accurate evaluation of multi-dimensional indicators such as diseases and insect pests, mechanical damage, skin defects, texture characteristics, and nutrient components. In particular, the combination with machine learning and deep learning algorithms has greatly improved the data processing capability and detection accuracy of spectroscopic detection, providing new solutions for the field of agricultural product quality detection and further accelerating the detection efficiency of agricultural product quality. Zhang et al. [7] explored the application of HSI technology combined with machine learning in pear quality assessment, and the accuracy of the predicted correlation coefficient could reach 0.99; Zhao et al. [8] reconstructed hyperspectral information from RGB (red, green, blue) images through deep learning algorithms, providing a new method for tomato quality monitoring. Benmouna et al. [9] studied the application of visible-near-infrared (Vis-NIR) spectroscopy in the maturity classification of Fuji apples, and the correct classification rate could reach 96.5% by combining it with the Convolutional Neural networks (CNN) model.

By sorting out the research progress in the field of spectroscopic detection, this paper systematically summarizes the application research of spectroscopic technology in agricultural product quality detection, covering various agricultural products from fresh produce, fruits to vegetables, from the basic principles of spectroscopic technology to practical applications, and from technical challenges to future prospects, in order to provide comprehensive technical references for researchers and practitioners in related fields.

## Overview of spectroscopic technology types

Spectroscopic detection technology is usually based on the principle of the optical properties of substances, such as absorption, scattering, and reflection of light with different wavelengths. By analyzing the spectral response of samples, it realizes the corresponding component analysis and quality assessment. The main types of spectroscopic detection technology include hyperspectral imaging (HSI), Vis-NIR, near-infrared spectroscopy (NIRS), mid-infrared spectroscopy (MIRS), Raman spectroscopy (Raman), X-ray computed tomography (X-ray CT), surface-enhanced Raman spectroscopy (SERS), fluorescence imaging, terahertz time-domain spectroscopy (THz-TDS), and electrochemical impedance spectroscopy (EIS).

### HSI technology

HSI technology combines the advantages of spectral analysis and image processing, and can simultaneously

acquire spectral and spatial information of samples. The data generated by the HSI system is a three-dimensional (3D) hypercube, including 2 spatial dimensions and one spectral dimension [10, 11]. This technology has high spectral resolution in the Vis-NIR range (400−1 000 nm), usually reaching several nanometers, and can capture subtle spectral characteristics of samples. In agricultural product detection, HSI has been widely used in fruit shape, quality evaluation, defect detection, and maturity judgment. By effectively combining spatial and spectral information, it realizes multi-dimensional data analysis [12, 13]. Meanwhile, in the application form of HSI, push-broom scanning is the most common acquisition method in agricultural applications [14]. The HSI system usually consists of a light source (such as a halogen lamp, light emitting diode (LED)), a light-splitting device (such as a prism, grating), a detector (such as a charge-coupled device (CCD), complementary metal-oxide-semiconductor (CMOS) camera), and a control computer [15].

#### *Detection principle of HSI technology*

HSI technology simultaneously acquires spatial and spectral information of target objects, forming a 2-dimensional (2D) image data cube (3D hypercube) containing hundreds of contiguous spectral bands [10]. The core principle of its detection lies in the following: when light irradiates the surface of agricultural products, substances absorb, reflect, or scatter light of specific wavelengths due to differences in chemical composition and physical structure. These optical responses can reflect the internal quality (e.g., sugar content, acidity, moisture) and external characteristics (e.g., damage, contamination) of agricultural products. The system typically consists of a light source, a spectral dispersion device (such as a grating or filter), a detector, and a data processing unit, with an operating wavelength range covering the visible to near-infrared regions [15].

#### *Analysis methods of HSI technology*

i) Data preprocessing: Includes spectral correction (removing background noise, dark current compensation), spatial calibration (adjusting image resolution and geometric distortion), and spectral preprocessing (e.g., normalization, derivative transformation, multiplicative scatter correction (MSC)) to enhance effective signals for identification.

ii) Feature extraction and dimensionality reduction: Methods such as principal component analysis (PCA), successive projections algorithm (SPA), or band selection are adopted to screen key wavelengths from massive data and reduce computational complexity.

iii) Modeling and classification: Combined with chemometric methods (e.g., partial least squares regression (PLSR), support vector machine (SVM)) or deep learning models, quantitative or qualitative relationships between spectral features and quality parameters are established to achieve quality evaluation and defect identification.

#### *Current dilemmas and limitations of HSI technology*

i) Large data volume and low processing efficiency: Hyperspectral images contain dozens to hundreds of bands with high data dimensionality, leading to significant storage, transmission, and computational costs that make it difficult to meet real-time detection requirements.

ii) High system cost and complexity: The high cost of precision optical components and customized system construction, coupled with the need for professional operators for maintenance, limits industrial application.

iii) Poor environmental sensitivity and stability: Detection results are easily interfered by environmental factors (e.g., temperature, light changes) and physical properties of samples (e.g., surface curvature, color), requiring frequent calibration to ensure accuracy.

### Vis-NIR spectroscopy technology

The detection principle of Vis-NIR spectroscopy technology is to analyze the absorption and reflection characteristics of substances in the wavelength range of 400−1 000 nm. This spectral region contains information on the overtone and combination frequencies of molecular bonds (such as O−H, C−H, and N−H bonds), which can effectively reflect the main chemical components of agricultural products [10]. At the same time, Vis-NIR spectroscopy technology has the advantages of fast detection speed and no need for sample pretreatment, making it very suitable for rapid non-destructive detection of agricultural product quality [16, 17].

#### *Detection principle of Vis-NIR spectroscopy technology*

Vis-NIR spectroscopy technology leverages the overtone and combination band vibrational absorption characteristics of chemical bonds (e.g., C−H, O−H, N−H) in organic molecules within specific wavelength ranges. By

measuring the absorption, reflection, or transmission intensity of light by samples, it acquires characteristic spectral signals related to their chemical compositions (e.g., moisture, protein, sugar content) and physical properties [10]. After the sample is irradiated by a light source, the spectrometer collects reflected or transmitted light signals, which are converted into spectral curves over a continuous wavelength range through photoelectric conversion. The position and intensity of absorption peaks in these curves exhibit a quantitative relationship with the content of analytes.

#### *Analysis methods of Vis-NIR spectroscopy technology*

i) Spectral preprocessing: MSC, standard normal variate (SNV), or derivative methods (e.g., first derivative) are employed to eliminate baseline drift and scattering interference, thereby enhancing effective signals.
ii) Characteristic wavelength selection: PCA, SPA, or genetic algorithm (GA) are used to reduce data dimensionality and screen key wavelengths with high correlation to target parameters.
iii) Modeling and validation: Quantitative models between spectra and quality parameters are established based on chemometric methods (e.g., PLSR, SVM). Model accuracy is evaluated via cross-validation and external validation sets using metrics such as the coefficient of determination ($R^2$) and root mean square error (RMSE).

#### *Current dilemmas and limitations of Vis-NIR spectroscopy technology*

i) Insufficient model generalization ability: Calibration models heavily rely on the variety, origin, and growth environment of training samples. Their accuracy declines significantly when applied across regions or varieties, necessitating frequent updates of calibration sets.
ii) Limited detection of low-concentration components: The sensitivity to trace components (e.g., vitamins, heavy metals) is relatively low. Weak signals and susceptibility to background interference make accurate quantification challenging.
iii) Sensitivity to environment and operation: Detection results are strongly influenced by temperature, humidity, sample particle size, and surface morphology. Strictly standardized operating procedures are required to ensure stable performance.

### NIRS technology

The working principle of NIRS technology is usually based on the overtone and combination frequency vibrations of groups such as C−H, O−H, and N−H in molecules, which mainly reflects the composition information of organic substances. This technology has the advantages of rapidity, non-destructiveness, and no need for sample pretreatment, and is widely used in the detection of physical and chemical indicators such as sweetness, hardness, and maturity of agricultural products [18]. Its advantages lie in relatively low equipment cost, fast acquisition speed, and good reflection of the chemical composition and structural information of samples, making it particularly suitable for the quantitative analysis of components such as moisture, protein, and fat in agricultural products [19–21].

#### *Detection principle of NIRS technology*

NIR spectroscopy technology is based on the principle of molecular vibrational spectroscopy in the quality detection of agricultural products. It achieves non-destructive and rapid detection of components such as moisture, protein, fat, and sugar by analyzing the absorption, reflection, or transmission characteristics of samples towards near-infrared light [18, 19]. Its core lies in acquiring spectral signals through the overtone and combination band vibrational absorption of chemical bonds (e.g., C−H, O−H, N−H) and establishing a quantitative relationship with target components.

#### *Analysis methods of NIRS technology*

i) Spectral collection and preprocessing: Diffuse reflectance or transmission mode is adopted to collect spectra. MSC, SNV, or derivative methods are used to eliminate noise and baseline drift.
ii) Characteristic wavelength selection: PCA, GA, or SPA are utilized to reduce data dimensionality and extract key wavelengths.
iii) Modeling and validation: Quantitative models are established based on PLSR, SVM, or artificial neural network (ANN). Model accuracy is evaluated via cross-validation using metrics such as $R^2$ and RMSE.

#### *Current dilemmas and limitations of NIRS technology*

i) High cost and technical threshold: The requirements for high-precision spectrometers and chemometric modeling have raised application costs, mak-

ing it difficult for small and medium-sized producers to popularize the technology.
ii) Insufficient real-time performance: Complex model calculations and hardware delays make it hard to meet the real-time detection requirements of high-speed production lines.
iii) Sensitivity to environment and operation: Detection results are significantly affected by temperature, humidity, sample particle size, and surface morphology, requiring strictly standardized operating procedures.

## MIRS technology

The working principle of MIRS technology involves the fundamental frequency vibration of molecules, which provides rich molecular structure information and is particularly suitable for the qualitative and quantitative analysis of organic compounds. MIRS is mainly used in component analysis, deterioration detection, and authenticity identification in agricultural product detection, and is widely used in the authenticity identification and quality assessment of products such as oils, dairy products, and honey [22].

### *Detection principle of MIRS technology*

MIR spectroscopy technology is based on the principle of molecular vibrational spectroscopy. It detects the fundamental frequency vibrational signals of chemical components in agricultural products by analyzing the characteristic absorption of mid-infrared light by samples, enabling non-destructive or micro-destructive analysis of quality indicators. Its core lies in the stretching and bending vibrations of chemical bonds (e.g., C=O, O−H, N−H), which form unique "fingerprint spectra" suitable for qualitative identification and quantitative analysis [22].

### *Analysis methods of MIRS technology*

i) Spectral collection and preprocessing: Fourier transform infrared (FTIR) spectroscopy is mainly adopted, combined with an attenuated total reflectance (ATR) accessory to handle high-moisture or inhomogeneous samples. Derivative methods or SNV are used to eliminate baseline interference.
ii) Dimensionality reduction and pattern recognition are performed using PCA. Quantitative models are established by combining PLSR or SVM, which are applied to the detection of indicators such as sugar content, organic acids, and acid value of oils and fats.
iii) Cluster analysis is used to achieve origin tracing, variety identification, and adulteration identification of agricultural products (e.g., sugar addition in fruit juices, oil adulteration).

### *Current dilemmas and limitations of MIRS technology*

i) Limited sample adaptability: High-moisture or inhomogeneous samples (e.g., fruits, vegetables, grains) are susceptible to scattering effects. Complex preprocessing (e.g., drying, grinding) is required, making it difficult to achieve truly non-destructive detection.
ii) Insufficient sensitivity and quantitative accuracy: The detection limit for low-concentration components (e.g., trace additives, toxins) is relatively high. Additionally, models rely on calibration with standard chemical methods, resulting in weak generalization ability across varieties.
iii) Complexity of spectral interpretation: Overlapping absorption peaks (e.g., the spectral range of carbohydrates and proteins) require in-depth analysis using chemometrics, leading to high costs for model maintenance and updates.

## Raman technology

The working principle of Raman technology is usually based on the inelastic scattering effect, which provides information on the vibration and rotation of molecules, complementing infrared spectroscopy. At the same time, it has the characteristics of fast detection speed and high accuracy [23–25]. This technology is insensitive to moisture, making it particularly suitable for the analysis of water-containing samples. It is mainly used in fields such as component analysis and quality evaluation in agricultural product detection [26, 27]. Meanwhile, this method can effectively distinguish ferns, monocotyledons, and dicotyledons with an accuracy of 91%, and the use of Raman to detect lignin characteristics has been proven to be an effective classification index [28].

### *Detection principle of Raman technology*

Raman technology is mainly based on the Raman scattering effect. It enables effective identification of components in agricultural products by analyzing the characteristic spectra generated from the inelastic scattering between laser light and sample molecules [23,

24]. Its core principle lies in the polarizability changes induced by molecular vibrations, making it particularly suitable for detecting carbon chain skeleton vibrations, aromatic compounds, and non-polar functional groups.

#### *Analysis methods of Raman technology*

i) Conventional Raman: Suitable for the detection of high-concentration components (e.g., oils, pigments) but is susceptible to fluorescence interference.
ii) SERS: Enhances signals through metal nanostructures, achieving a significant improvement in sensitivity. It can be applied for the trace detection of pesticide residues (e.g., organophosphates on fruit surfaces) and toxins.
iii) Spatially offset Raman spectroscopy (SORS): Can effectively penetrate the surface layer to obtain deep tissue information, making it suitable for the internal quality analysis of thick-skinned fruits and vegetables (e.g., sugar content of watermelons).

#### *Current dilemmas and limitations of Raman technology*

i) Fluorescence interference and signal-to-noise ratio constraints: Pigments and phenolic compounds in agricultural products tend to produce strong fluorescence backgrounds that mask Raman signals. This issue is particularly prominent when excited by visible light lasers, requiring the use of near-infrared lasers or quenching technologies for mitigation.
ii) Insufficient sensitivity and quantitative stability: The detection of trace components (e.g., pesticides at concentrations < 0.1 mg/kg) relies on SERS enhancement. However, nano-substrates suffer from poor reproducibility and are easily interfered by sample matrices.
iii) Cost and portability bottlenecks: High-resolution spectrometers and stable laser sources are expensive. Although portable devices are available, there is a gap in accuracy compared to laboratory-grade instruments.

### X-ray CT technology

The working principle of X-ray CT technology usually involves measuring the absorption characteristics of materials to X-rays, which can obtain 2D or 3D information on the internal structure of samples [29]. This technology is mainly used in internal structure observation, defect detection, and density measurement in agricultural product detection.

#### *Detection principle of X-ray CT technology*

X-ray CT technology is based on the principles of X-ray attenuation characteristics and Radon transform. It reconstructs 2D slices and 3D models of the internal structure of objects through multi-angle projection data, enabling non-destructive detection of the internal quality of agricultural products [29]. Its core lies in generating grayscale images based on differences in X-ray attenuation coefficients among different substances (e.g., varying attenuation characteristics of moisture, fat, and pore structures), thereby quantifying internal features.

#### *Analysis methods of X-ray CT technology*

i) Image reconstruction and processing: Filtered back projection (FBP) or iterative algorithms are employed to reconstruct CT images. Gaussian filtering and median filtering are used to reduce noise, and threshold segmentation is performed based on grayscale histograms to extract target regions. 3D modeling can visualize pore distribution, tissue density gradients (e.g., fruit flesh sugar content, grain endosperm structure), and defective areas (e.g., insect infestation, mechanical damage).
ii) Quantitative and qualitative analysis: It can effectively evaluate internal quality by establishing linear models between CT values and physicochemical parameters (e.g., moisture, acidity, soluble solids content (SSC)). It enables microstructure observation to quantify porosity, cell wall thickness, and fat distribution (e.g., evaluation of meat marbling) as well as ice crystal formation processes (analysis of frozen agricultural products). Additionally, it can measure mechanical properties, and predict compressive strength and hardness (e.g., correlation between fruit ripeness and texture) in combination with finite element analysis.

#### *Current dilemmas and limitations of X-ray CT technology*

i) High complexity of image processing: The superposition of multiple components in agricultural products leads to cross-interference of attenuation coefficients (e.g., high-moisture backgrounds in fruits and vegetables masking minor defects). Complex algorithms are required for signal separation, resulting in time-consuming computation and insufficient model generalization ability.
ii) Cost and equipment constraints: CT systems are expensive (especially high-energy industrial CT equipment). Moreover, they only support cross-sec-

tional scanning, leading to poor adaptability to agricultural products with irregular shapes (e.g., curved root vegetables).

## SERS technology

The working principle of SERS usually involves enhancing the Raman signal through the plasmon resonance effect of nanometallic structures (such as Au, Ag nanoparticles (NPs)), with a sensitivity reaching the single-molecule level. The substrate design of this method is the key, including metal nanoparticles, core–shell structures, and mixed materials (such as ZnO/Ag composites). This method is suitable for the detection of trace pollutants (such as pesticides, toxins) [30].

### *Detection principle of SERS technology*

SERS is a high-sensitivity detection technology that significantly amplifies the Raman signals of target molecules based on the surface plasmon resonance effect of nanostructured metal materials (e.g., gold, silver, copper). The core principle of its detection mainly relies on the electromagnetic field enhancement mechanism and the chemical enhancement mechanism [30].

### *Analysis methods of SERS technology*

i) Substrate preparation and optimization: Noble metal nanomaterials (e.g., silver nanorod arrays, gold nanostars) are used to construct high-density "hot spots" for improving detection sensitivity. Surface modification (e.g., cyclodextrins, aptamers, molecularly imprinted polymers) is adopted to enhance the selective capture capability for specific targets (e.g., pesticides, antibiotics).
ii) Detection modes: Mainly divided into direct detection and indirect detection. The former utilizes the inherent Raman fingerprint spectra of target molecules for qualitative and quantitative analysis (e.g., pesticide residues, illegal additives). The latter combines labeling molecules (e.g., dyes, DNA probes) with specific recognition elements to achieve highly selective detection (e.g., mycotoxins, heavy metal ions).
iii) Data processing and modeling: Algorithms such as wavelet denoising and background subtraction are applied to process raw spectra. Quantitative models are established in combination with chemometrics (e.g., PCA, PLSR). Machine learning and multivariate analysis can also be employed to facilitate the simultaneous identification of multiple components in complex matrices.

### *Current dilemmas and limitations of SERS technology*

i) Insufficient substrate stability and reproducibility: Colloidal nanoparticles are prone to aggregation and precipitation, solid-phase substrates exhibit large batch-to-batch variations, and noble metal materials are susceptible to oxidation and deactivation.
ii) Components such as pigments and proteins in agricultural products competitively adsorb on the substrate surface, masking the signals of target analytes. Complex pretreatment is required, which increases operational costs.
iii) The lack of unified standards for evaluating substrate performance and a Raman spectral database of agricultural product components limits the model generalization ability and industrial application.

## Fluorescence imaging technology

Fluorescence imaging usually excites endogenous fluorescent substances (such as chlorophyll, polyphenols) in samples through ultraviolet or visible light, and captures the emission spectrum to detect surface damage and spoilage [31]. Its core principle is that specific substances (fluorescent probes/fluorescent molecules) absorb energy and jump to the excited state under the irradiation of external excitation light (such as ultraviolet, visible light), and then release energy in the form of fluorescence (light of a specific wavelength) when returning from the excited state to the ground state. By detecting the intensity, position, or spectral information of the fluorescence signal, it realizes the imaging and quantitative analysis of target molecules, cells, or tissues.

### *Detection principle of fluorescence imaging technology*

Fluorescence imaging spectroscopy realizes quality detection by capturing the fluorescence signals emitted by agricultural products under excitation by specific light, combined with image and spectral analysis. Its core principle is as follows: when agricultural products are irradiated with ultraviolet or visible light, inherent fluorescent substances (e.g., chlorophyll, polyphenols, vitamins) absorb light energy and emit fluorescence at specific wavelengths. Quality parameters can be inferred by analyzing fluorescence intensity, spectral shape, and spatial distribution [31].

### *Analysis methods of fluorescence imaging technology*

i) Fluorescence generation mechanism: When substances are excited by light of specific wavelengths,

electrons transition to higher energy states and release fluorescence when returning to the ground state. Chlorophyll, flavonoids, lipids, and other components in agricultural products can all produce characteristic fluorescence, whose intensity is directly related to the substance concentration and metabolic state.

ii) Fusion of imaging and spectroscopy: Hyperspectral fluorescence imaging synchronously acquires spatial details and continuous spectral data. It can not only locate surface defects (e.g., bruises, mold spots) but also quantify internal components (e.g., sugar content, acidity). Algorithms such as PCA and partial least squares (PLS) are used to extract characteristic spectra and establish quality prediction models (e.g., apple sugar content, citrus rot degree).

#### *Current dilemmas and limitations of fluorescence imaging technology*

i) Poor stability of fluorescence signals: Ambient light interference, temperature fluctuations, and sample surface inhomogeneity lead to large variations in fluorescence intensity and low signal-to-noise ratio (especially for fruits and vegetables with high water content), requiring frequent calibration.

ii) Complex matrix interference: Spectral peak overlap of multiple fluorescent substances in agricultural products (e.g., chlorophyll and carotenoids) makes it difficult to resolve signals of specific targets, relying on complex chemometric models.

### THz-TDS technology

Terahertz waves are located between microwaves and infrared rays (0.1 – 10 THz), and are sensitive to polar molecules, especially water molecules. At the same time, they have the property of penetrating non-polar materials [32]. THz-TDS technology obtains optical parameters such as the absorption coefficient and refractive index of samples by measuring the change in the electric field intensity of terahertz pulses before and after passing through the samples. Terahertz technology is suitable for internal defect detection, moisture content analysis, and pesticide residue detection in agricultural product detection [33, 34].

#### *Detection principle of THz-TDS technology*

THz-TDS technology enables component detection and quality evaluation by leveraging the interactions between terahertz waves and molecular bond vibrations, rotations, as well as lattice vibrations in agricultural products [32]. Its core principle is as follows: after irradiating the sample with terahertz pulses, optical parameters such as the refractive index and absorption coefficient of the sample are extracted by measuring changes in electric field amplitude and phase, thereby inverting the concentration and distribution of components including moisture, protein, and carbohydrates.

#### *Analysis methods of THz-TDS technology*

i) Feature extraction and modeling: Algorithms such as PCA and PLSR are used for dimensionality reduction and characteristic frequency screening. Machine learning methods like SVM are combined to conduct classification and identification of fruit and vegetable quality.

ii) Imaging fusion technology: The combination of terahertz imaging and spectroscopy generates spatial distribution maps of components (e.g., corn endosperm structure, foreign object localization in agricultural products), realizing simultaneous evaluation of morphology and composition.

#### *Current dilemmas and limitations of THz-TDS technology*

i) Significant moisture interference: Agricultural products with high moisture content strongly absorb terahertz waves, leading to severe signal attenuation and limited penetration depth (usually < 1 mm), which makes it difficult to detect internal defects.

ii) Insufficient sensitivity and resolution: Characteristic peaks of low-concentration components (e.g., pesticide residues) are easily masked by matrix signals, relying on complex chemometric models. Additionally, the signal-to-noise ratio of the equipment is low.

### EIS technology

The working principle of EIS technology is to indirectly reflect the internal components (such as moisture, ion concentration) of biological tissues by measuring their electrical response. It is suitable for the detection of fruit and vegetable maturity, meat freshness, etc., mainly including food freshness evaluation (eg., meat, fruits and vegetables), food packaging integrity detection, and emulsion stability analysis [35].

#### *Detection principle of EIS technology*

EIS reflects quality-related parameters such as internal tissue structure, moisture content, and cell integrity by applying alternating current (AC) signals of different frequencies to agricultural products and measuring their impedance responses (including the relationships

between amplitude, phase, and frequency). Its principle is based on the electrical properties of biological tissues, which is effectively realized through cell membrane characteristics and the establishment of electrical models [35].

#### *Analysis methods of EIS technology*

i) Establish machine learning models: Combine multivariate statistics (e.g., PCA) or ANN to predict quality from full-spectrum data, such as in fields like fish freshness classification and rice moisture prediction.
ii) Characteristic frequency extraction: Select impedance values or phase angles at specific frequencies to correlate directly with quality parameters.

#### *Current dilemmas and limitations of EIS technology*

i) Interference from individual differences in agricultural products: Variations in size, shape, location, and physiological state (e.g., variety, growth conditions) of agricultural products lead to significant differences in impedance responses, making it difficult to establish universal models.
ii) Influence of environmental sensitivity: Environmental factors such as temperature and humidity significantly affect measurement accuracy. Strict temperature control or compensation algorithms are required, increasing the difficulty of on-site application.
iii) Limitation of application scope: Current research focuses on a few categories (e.g., rice, beef, tomatoes), while systematic research on diversified agricultural products (e.g., leafy vegetables, nuts, aquatic products) is insufficient, limiting technological popularization.

The summary and comparison of specific spectral characteristics are shown in Table 1.

## Analysis of spectroscopic detection process

When spectroscopic detection technology is applied to the quality detection of agricultural products, it is usually divided into 4 steps: i) Ensuring the stability of sample and instrument status; ii) Spectral acquisition and reference value determination; iii) Data processing:

**Table 1** Summary and comparison of the characteristics of main spectral technologies

| No. | Technology Type | Spectral range | Resolution | Main applications | Advantages | Limitations | References |
|---|---|---|---|---|---|---|---|
| 1 | HSI | 200–2 500 nm | 2–5 nm | Quality evaluation, defect detection | Combination of spatial and spectral information | Large data volume, complex processing | [12, 13] |
| 2 | Vis-NIR | 400–1 000 nm | 0.5–5.0 nm | Rapid non-destructive detection of agricultural products | Fast speed, no pretreatment required | High hardware cost, large site limitations | [10, 16, 17] |
| 3 | NIRS | 780–2 500 nm | 1–10 nm | Quantitative component analysis | Rapid, non-destructive | Relatively low sensitivity | [18–21] |
| 4 | MIRS | 2 500–25 000 nm | 1–4 $cm^{-1}$ | Structural analysis, authenticity identification | Rich information | Sensitive to moisture | [22, 36] |
| 5 | Raman | 200–4 000 $cm^{-1}$ | 1–2 $cm^{-1}$ | Component analysis, quality evaluation | Insensitive to moisture | Weak signal, fluorescence interference | [23–28] |
| 6 | X-ray CT | 10 pm–10 nm | 5–50 μm | Internal structure observation | 3D internal structure | Expensive equipment, radiation issues | [29] |
| 7 | SERS | 50–4 000 $cm^{-1}$ | 2–5 $cm^{-1}$ | Pesticide residue and toxin detection | Trace detection, molecular fingerprinting | Poor substrate stability, high cost | [30] |
| 8 | Fluorescence Imaging | 200–1 700 nm | 1–5 nm | Surface damage and oxidation state detection | High sensitivity, specificity | Dependent on fluorescent substances | [31, 37] |
| 9 | THz-TDS | 0.1–10.0 THz | 1–100 GHz | Internal defect detection, moisture analysis, pesticide residue detection | Penetrates non-polar materials, sensitive to water | Low resolution, expensive equipment | [32–34] |
| 10 | EIS | 0.01 mHz–100 MHz | 1–20 Hz | Fruit and vegetable maturity, meat freshness detection | High sensitivity, non-destructive detection, no labeling required | High experimental complexity, time-consuming low-frequency testing | [35] |

eliminating noise and establishing prediction models; iiii) Actual detection and result analysis (Fig. 2). Among them, the latter 3 steps are particularly important.

### Spectral acquisition modes

When light irradiates a sample, part of the light is directly reflected by the sample surface, and part of the light penetrates the sample tissue and is absorbed and scattered inside. According to the propagation path of light and the detection method, spectral acquisition is mainly divided into 3 modes: reflectance mode, transmittance mode, and interaction mode [38].

i) In reflectance mode, the light source and detector are located on the same side of the sample, mainly collecting the reflected light information on the sample surface.
ii) In transmittance mode, the light source and detector are located on opposite sides of the sample, detecting the signal of light after penetrating the sample.
iii) In interaction mode, a light barrier is set between the light source and the detector to prevent the interference of surface reflected light and obtain deeper information.

### Spectral data pretreatment

Spectral data acquisition is easily affected by factors such as environmental noise, instrument response, and sample status, so pretreatment is required to improve the signal-to-noise ratio and model performance. Common pretreatment methods include: 1) Savitzky-Golay smoothing algorithm: eliminating random noise through polynomial fitting to improve the signal-to-noise ratio [39, 40]. 2) Derivative processing: using first-order and second-order derivatives to eliminate baseline drift and background interference, and enhance spectral characteristics. 3) SNV transformation: effectively eliminating the influence of particle size and surface reflection. 4) MSC: effectively eliminating the influence of light scattering and enhancing absorption characteristics related to components [41].

### Variable selection methods

Spectral data usually contains a large amount of redundant information. The use of variable selection methods can improve model efficiency and prediction ability. Common methods mainly include: 1) SPA: selecting variable groups with minimal collinearity. 2) Competitive adaptive reweighted sampling (CARS): selecting information-rich variables based on PLS regression coefficients. 3) GA: optimizing variable subsets by simulating the natural selection process.

### Qualitative and quantitative modeling methods

In the quality detection of agricultural products using spectroscopy, algorithms and modeling methods are usually combined to effectively improve detection accuracy. Common methods mainly include: 1) PLSR: the most commonly used linear regression method, suitable for modeling between spectra and component contents, and widely used in quantitative prediction [42]; 2) Least squares SVM (LS-SVM): suitable for modeling small-sample and non-linear problems [43]. 3) ANN: capable of handling complex non-linear relationships [44, 45]. 4) Deep learning algorithms: extracting deep-level features through

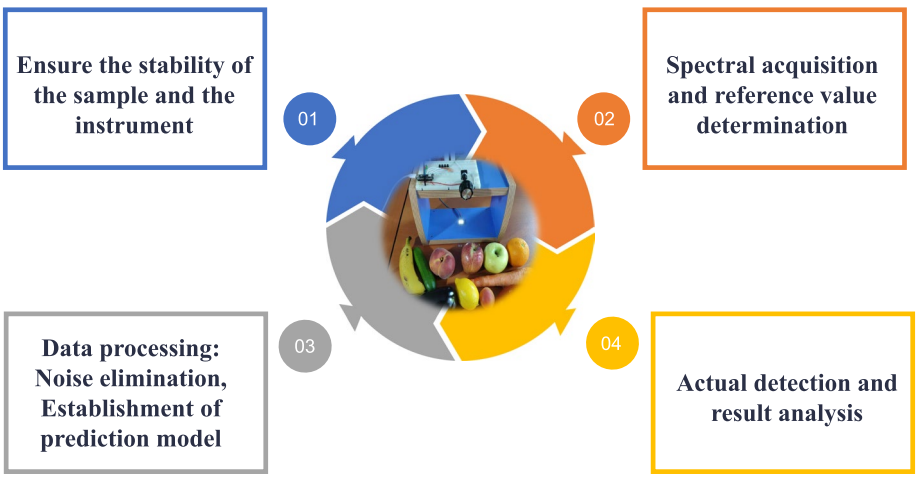



**Fig. 2** Operation process of spectroscopic detection for agricultural products

multi-layer neural networks. Deep learning (such as combining with CNN, GAN) enhances feature extraction and model robustness for agricultural products [46]. 5) Random forest (RF): capable of resisting overfitting and evaluating feature importance, suitable for high-dimensional data and feature selection applications. 6) PCA: capable of effective dimensionality reduction, visualization, and anomaly detection, suitable for data exploration, sample classification, and other detection fields.

### Model evaluation indicators

The widely used quantitative models of evaluation indicators include correlation coefficient ($R$), $R^2$, root mean square error of calibration (RMSEC), root mean square error of prediction (RMSEP), and relative percent difference (RPD). Generally speaking, the closer $R^2$ is to 1, and the smaller RMSEC and RMSEP are, the better the model performance. RPD > 2 indicates that the model has good prediction ability, and RPD > 3 indicates that the model has excellent prediction ability. The widely used qualitative models of evaluation indicators include overall accuracy (Acc), F1-score (F1), precision (P), and recall (R) [47]. Acc reflects the overall classification accuracy; F1-score balances precision and recall, indicating their comprehensive trade-off; precision denotes the proportion of samples predicted as positive that are actually positive; recall represents the proportion of actual positive samples that are correctly identified by the model. The specific evaluation indicators and standards are shown in Table 2.

## Application of spectroscopic technology in agricultural product quality detection

### External quality evaluation

Spectroscopic technology is also widely used in the external quality detection of agricultural products. In the detection of external defects and damage, HSI and fluorescence imaging are usually used to efficiently detect surface defects. HSI and fluorescence imaging are used to identify and detect orange rot and mechanical damage with an accuracy > 90% [31]. In dealing with the color and texture of agricultural product surfaces, Vis-NIR is usually used to predict color parameters (such as brightness ($L^*$), redness ($a^*$), yellowness ($b^*$)) and texture. Zhang et al. [48] reviewed the influence of color variability of agricultural products on models, which requires combining spectroscopic technology for band ratio or normalization processing. Elsayed et al. [49] conducted detection on the firmness and maturity of strawberries and mangoes by combining spectral technology and machine learning, ultimately achieving $R^2$ values ranging from 0.92 to 1.00, which enables accurate prediction of the relevant properties of mango and strawberry fruits. Zhang et al. [50] pointed out in their review that for different pear varieties, adopting methods such as data fusion and advanced machine learning can effectively identify the external varieties of pears as well as their firmness and maturity. Munera-Picazo et al. [51] performed external quality evaluation of intact persimmons using NIRS. By integrating algorithms and models, they achieved a success rate of over 92% in effectively identifying the maturity of persimmons and could accurately detect the firmness of persimmons.

**Table 2** Common evaluation indicators and standards for spectral modeling

| No. | Evaluation indicator | Calculation formula | Evaluation standard | Excellent range | Acceptable range | References |
|---|---|---|---|---|---|---|
| 1 | $R^2$ | 1 − SSE/SSR | Closer to 1 is better | > 0.90 | 0.75–0.90 | [7, 8] |
| 2 | RMSEC | $\sqrt{\frac{\sum(y_e-y_r)^2}{n}}$ | Smaller is better | < 0.1 SD | < 0.2 SD | [7, 8] |
| 3 | RMSEP | $\sqrt{\frac{\sum(y_p-y_r)^2}{n}}$ | Smaller is better | < 0.1 SD | < 0.2 SD | [7, 8] |
| 4 | RPD | SD/RMSEP | Smaller is better | > 3 | 2–3 | [7, 8] |
| 5 | Acc | $\frac{TP+TN}{TP+TN+FP+FN}$ | Overall classification accuracy | > 0.90 | >0.80 | [47] |
| 6 | F1 | $\frac{2P\times R}{P+R}$ | Suitable for imbalanced data | > 0.90 | 0.80–0.90 | [47] |
| 7 | P | $\frac{TP}{TP+FP}$ | The proportion of samples predicted as positive that are actually positive | > 0.90 | 0.80–0.90 | [47] |
| 8 | R | $\frac{TP}{TP+FN}$ | The proportion of actual positive samples that are correctly identified | > 0.90 | 0.80–0.90 | [47] |

### Internal quality evaluation

Spectroscopic technology shows great potential in the internal quality evaluation of agricultural products. In the detection of SSC, Vis-NIR is generally used to predict the content of SSC. Nicolai et al. [52] used this technology to detect the apple SSC with a predicted RMSEP of 0.5–1.0°Brix. Amoriello et al. [53] used NIR scattering to measure kiwifruit hardness when detecting the hardness and dry matter (DM) indicators of agricultural products, and the measured hardness $R^2$ was 0.72; they used HSI to detect strawberry parameters, and the measured strawberry SSC prediction accuracy $R^2$ was 0.981, strawberry DM prediction accuracy $R^2$ was 0.942, and strawberry hardness prediction accuracy $R^2$ was 0.951. Zhang et al. [7] used HSI technology combined with machine learning algorithms to conduct a comprehensive quality evaluation of 6 pear varieties, and the prediction accuracy of the calibration set and test set exceeded 99%. X-ray CT technology has unique advantages in internal structure detection. Du et al. [29] reviewed the application of X-ray CT in agricultural product quality detection, including various products such as wheat, corn, rice, apples, pears, beef, and mutton. This technology can obtain 3D information on the internal structure of samples, which is used for the detection of porosity, density, and internal defects. The final study showed that the X-ray CT values had a good correlation with the sample density. Tang et al. [54] utilized HSI technology to collect spectral data of bruised apples under four different exposure durations. They preprocessed the spectral data using PCA, ultimately achieving a bruise detection accuracy of 97.33%. Liu et al. [55] employed HSI combined with chemometric methods to detect defects in strawberries, obtaining an area under the curve (AUC) of 0.923, which enables effective identification of abnormalities and defects in strawberries. Munera et al. [56] applied hyperspectral technology to detect internal stone cracking in nectarines. By adopting PLS-discriminant analysis (PLS-DA), they effectively improved detection efficiency, with an accuracy rate of 94.7%.

### Nutrient component analysis

Spectroscopic technology has achieved significant results in the nutrient component analysis of agricultural products. In peanut quality detection, Yang et al. [37] reviewed the application of spectroscopic technology in the detection of protein, moisture, and oil content. By establishing a near-infrared determination model for peanut seed oil content, the correlation coefficient between the predicted value and the chemical determination value of the verification sample was extremely high; at the same time, a calibration model for oleic acid and linoleic acid in peanut kernels could also be established, which could accurately and reliably predict the actual values; a Fourier transform near-infrared spectrometer was used to quantitatively analyze the contents of main fatty acids such as oleic acid, linoleic acid, and palmitic acid in peanut seeds, providing technical support for high-oleic acid peanut breeding. In fish oil quality detection, Wang et al. [57] reviewed the application research of spectroscopic technology in fish oil quality detection. A PLS quantitative prediction model was established using NIRS technology to accurately predict the content of various nutrients in salmon oil; NIR technology was also used to quickly detect the contents of eicosapentaenoic acid (EPA), docosahexaenoic acid (DHA), phospholipids, astaxanthin, and other substances in krill oil. Chen et al. [58] analyzed the nutritional components of pomelos by combining near-infrared hyperspectral technology with PLS technology. Ultimately, a RMSEP of 1.08% was achieved for the detection of sugar content and VC content in pomelos, demonstrating that this method can efficiently determine the nutritional component contents of pomelos. Yang et al. [59] established a prediction model for the soluble sugar content of Chinese chestnuts using hyperspectral technology, obtaining a RMSE of 0.94. This proves that NIRS technology has the potential to serve as a rapid and non-destructive method for predicting the soluble sugar content of Chinese chestnuts, which is conducive to quality control and sorting in the food industry. Elmetwall et al. [60] integrated hyperspectral technology with RGB technology and evaluated the nutritional quality of tomatoes and citrus fruits via machine learning. $R^2$ of 0.96 was obtained, indicating that this approach is scientifically feasible.

### Freshness and safety detection

Among the quality indicators of agricultural products such as fresh produce, freshness is a key indicator concerned by consumers. Li et al. [61] reviewed the application of spectroscopic technology in aquatic product quality detection, including freshness detection, harmful substance residue detection, harmful microorganism content detection, quality grading, and adulteration analysis. HSI technology combined with deep learning algorithms can be used to predict the total volatile basic nitrogen (TVB-N) content of Pacific white shrimp during cold storage; NIRS technology and computer vision technology can be combined to predict the TVB-N content of fish; the hyperspectral band of 400−1 000 nm can be used to measure the *Escherichia coli* colony in grass carp meat, ensuring the overall quality of aquatic products. At the same time, NIR-HSI technology was used to quickly and non-destructively detect the TVB-N content in chicken, effectively preventing spoiled chicken from entering the market [36]. Johnson et al. [62] reviewed the application

of infrared spectroscopy in the detection of bioactivity in food, demonstrating that spectral technology can effectively detect food freshness. Liakos et al. [63] summarized that the combination of hyperspectral detection technology and machine learning technology can effectively identify food freshness, as well as the defects and safety of agricultural products, meat products, and dairy products. Su et al. [64] conducted freshness detection of potatoes using spectral technology. Verified and summarized through experiments, the final freshness detection rate reached 98%, indicating that spectral technology is of great significance for the freshness detection of agricultural products.

#### Adulteration identification and authenticity analysis

The adulteration of agricultural products is a global challenge, and spectroscopic technology provides an effective means for adulteration identification. In the review by Ma et al. [65], it was explained that Vis-NIR spectroscopy was used to establish a prediction model for the content of vegetable oil adulterated in fish oil, and a prediction model was established by combining spectroscopic technology with the PLS algorithm. The results showed that the prediction model had good robustness in distinguishing oil quality; in terms of brand identification, short-wave NIRS, long-wave NIRS, and MIRS technologies were used to collect the spectral characteristics of 7 different brands of fish oil, and PLS-DA and LS-SVM were used to establish discriminant models. The PLS-DA model and LS-SVM model based on long-wave NIRS achieved the highest identification accuracy, with both the modeling set and prediction set identification accuracy reaching 100%. In the review by Pereira et al. [66], NIR can effectively distinguish original milk from adulterated milk added with pseudo-protein and thickeners; at the same time, both NIR and MIR technologies can detect melamine residues in milk, and even harmful components in milk powder products can be detected. Feng et al. [67] used NIR and HSI technologies to classify different varieties of rice, wheat, and corn, and the detection accuracy exceeded 90%. Zeng et al. [68] reviewed the research on food quality discrimination using NIRS, which can be applied to the authenticity detection of honey, flour, and olive oil, achieving favorable discrimination results. Qin et al. [69] summarized the research on Raman in food safety and quality detection, which can effectively determine the melamine concentration in milk powder and facilitate effective further screening of milk powder quality. The applications of other spectroscopic detection technologies in agricultural products are shown in Table 3.

Some actual scenarios of the above spectroscopic application fields are shown in Fig. 3.

## Technical challenges and limitations

Although spectroscopic technology has achieved significant progress in agricultural product quality detection, the current related research field still faces some technical challenges and limitations:

1) Issue of model universality: Agricultural products of different varieties, producing areas, and growth environments have different spectral characteristics, resulting in poor universality of established models. Models established for a specific variety or region are often difficult to be directly applied to other varieties or regions, requiring re-calibration and optimization [100].
2) High equipment cost: High-precision spectroscopic instruments are expensive [101], especially HSI and X-ray CT equipment, and their large size limits their application in on-site and online detection [102]. Although portable equipment has lower costs, its accuracy and stability are often inferior to desktop equipment.
3) Influence of environmental interference: Environmental factors such as light conditions, temperature, and humidity have a significant impact on spectral measurement results, requiring measurements to be carried out under standardized conditions [103, 104]. In actual production environments, these factors are difficult to fully control, affecting the stability of detection results.
4) Complexity of data processing: Spectral data volume is large, especially hyperspectral image data. Processing and analysis require professional knowledge and skills. HIS generates a large amount of data sets during operation, requiring further advanced processing [105]. The process of feature extraction and model optimization is complex, which has high requirements for the professional quality of operators.
5) Limitations of detection accuracy and multi-technology integration: For some low-content components or complex matrix samples, the detection accuracy of spectroscopic technology is still inferior to traditional chemical methods. Multi-dimensional quality evaluation can be realized by combining technologies such as HSI, RS, and ML [106, 107], thereby improving detection accuracy in special cases.

## Future prospects

As an important means of agricultural product quality detection, spectroscopic detection technology has achieved significant research progress and practical application results. From HSI to Vis-NIR, from MIRS to Raman, and to X-ray CT technology, combined with

**Table 3** Applications of spectroscopic detection in agricultural products

| Application direction | Main technologies and models | Detection object | Key band parameters or methods | Performance indicators | References |
|---|---|---|---|---|---|
| External quality evaluation | FT-NIR + HSI | Kiwifruit DM, color | Band: 400–1 000 nm | $R^2$ = 0.854–0.914 | [70] |
| | SVM + HSI | Apple surface defect detection | Band: 400–700 nm | Accuracy = 85% | [71] |
| | PLS + HSI | Apple DM | Band: 400–1 100 nm | $R^2$ = 0.83, RPD = 2.39 | [72] |
| | CNN + HSI | Strawberry maturity | Band: 400–1 000 nm | Accuracy = 84% | [73] |
| | ANN + HSI | Mango maturity | Band: 450–1 000 nm | $R^2$ = 0.92–1.00 | [49] |
| | NIRS | Pear hardness | Band: 780–1 000 nm | $R^2$ = 0.974 | [50] |
| | MLR + NIR HSI | Potato DM | Band: 700–1 050 nm | $R^2$ = 0.96 | [74] |
| | PLSR + VIS-NIR HSI | Persimmon hardness | Band: 400–1 000 nm | $R^2$ = 0.80, RPD = 1.86 | [51] |
| | HSI | Persimmon light penetration depth | Band: 610–1 050 nm | Depth ≤ 2 mm | [75] |
| | SVM + HSI | Tomato drought stress | Band: 400–1 000 nm | $R^2$ = 0.959 0 | [76] |
| Internal quality evaluation | SVR + HSI | Lycopene in tomatoes | Band: 400–1 000 nm | $R^2$ = 0.965 2 | [77] |
| | HSI + SNV/MSC | Apple pH | Band: 400–1 100 nm | $R^2$ = 0.989 9 | [78] |
| | HSI | Apple SSC, DMC | Wavelength: 670 nm (chlorophyll absorption peak) | $R^2$ = 0.93 | [79] |
| | HSI+RF | Apple TSS | Band: 580–900 nm | $R^2$ = 0.73 | [80] |
| | HSI+ResNet/PWCF (residual network, piecewise nonlinear curve fitting) | Apple internal bruising | Band: 386–1 016 nm | Accuracy = 97.33% | [54] |
| | HSI | Strawberry bruise detection | Band: 1 000–1 600 nm | Accuracy = 92.3% | [55] |
| | HSI + PCA | Persimmon damage detection | Band: 780–2 526 nm | Accuracy = 90.8% | [81] |
| | Raman | Strawberry SSC, anthocyanins | Wavelength: 1 450 nm (water absorption band) | $R^2$ = 0.88–0.92 | [82] |
| | SNV-SPA-LS-SVM + HSI | Cerasus humilis SSC | Band: 895–1 700 nm | Rp = 0.8526, RMSEP = 0.970 3 | [83] |
| | MASS-Boss-extreme learning machines (ELM) + HSI/FSI | Kiwifruit SSC | Band: 400–1 000 nm | $R^2$ = 0.889 4 | [84] |
| | Con1dResNet + HSI | Cherry tomato SSC | Band: 400–1 000 nm | $R^2$ = 0.901 | [85] |
| | RF + HSI | Peach SSC | Band: 908.1–1 735.6 nm | $R^2$ = 0.9184, RMSE = 0.669 3 | [86] |
| | PLS-DA + HSI | Nectarine split pit detection | Band: 400–1 000 nm | Accuracy = 94.7% | [56] |
| Nutrient component analysis | CNN+HSI | Tomato nitrogen content | Band: 400–1 100 nm | Correct classification rate (CCR) = 97.33% | [87] |
| | Radial basis function (RBF)-PLS + HSI | Pomelo sugar, VC | Band: 1 000–2 500 nm | RMSEP = 1.08% | [58] |
| | PLS + HSI | Chinese chestnut soluble sugar content | Band: 400–1 000 nm | $R^2$ = 0.66, RMSE = 0.94 | [59] |
| | RGB + HSI | Citrus carotenoids | Band: 300–1 150 nm | $R^2$ = 0.96 | [60] |
| | SVR + HSI | Apple leaf nitrogen content | Band: 400–1 000 nm | $R^2$ = 0.739 | [88] |
| | PLSR + HSI | Avocado unsaturated fatty acid (UFA):saturated fatty acid (SFA) | Band: 400–1 000 nm | $R^2$ = 0.79, RPD = 2.06 | [89] |
| Freshness and safety detection | PLS + NIRS | Rice seed viability | Band: 750–25 000 nm | $R^2$ = 0.89–0.91 | [62] |
| | PLSR + NIRS | Beef moisture content | Band: 400–1 000 nm | $R^2$ = 0.849 | [35] |
| | CNN + NIRS | Egg freshness | Band: 400–1 000 nm | Accuracy = 98.73% | [63] |

**Table 3** (continued)

| Application direction | Main technologies and models | Detection object | Key band parameters or methods | Performance indicators | References |
|---|---|---|---|---|---|
| | PCA + FT-NIRS | Coffee quality and variety identification | Band: 4000–8 000 $cm^{-1}$ | Accuracy = 100% | [90] |
| | SVM + HSI | Apple disease classification | Band: 400–1 000 nm | Accuracy = 81.8% | [91] |
| | HSI + LS-SVM/SG-SNV | Wolfberry drying quality classification | Band: 400.680–1 001.612 nm | Accuracy = 96.66% | [92] |
| | CNN + HSI | Banana quality grade classification | Mask R-CNN | Accuracy = 98.4% | [93] |
| | Quantitative descriptive analysis (QDA) + HSI | Mango anthracnose | Band: 450–1 000 nm | Accuracy = 90.9% | [94] |
| | THz + K-Nearest Neighbor (KNN) | Insect foreign bodies in tea | Frequency: 0.4–1.4 THz | Accuracy = 98% | [95] |
| | Vis-NIR + PLSR | Potato freshness | Band: 450–1 000 nm | Accuracy = 98% | [64] |
| Adulteration identification and authenticity analysis | NIRS + SVM | Honey adulteration identification | Band: 210–400 nm | Accuracy = 84.44% | [68] |
| | Raman | Milk melamine | Band: 0–2 600 $cm^{-1}$ | Detection limit 0.2%–10% | [69] |
| | Raman | Bone residue in chicken mince | Band: 600–1 800 $cm^{-1}$, exposure time 10 s | Root Mean Square error of cross-validation (RMSECV) = 0.59% | [96] |
| | HSI + PCA | Rice quality identification | Band: 400–1 000 nm | Accuracy = 96% | [63] |
| | PLSR + NIRS | Black pepper adulteration identification | Band: 250–500 nm | Error < 5% | [90] |
| | Fluorescence spectroscopy | White wine purity | Band: 200–800 nm | Accuracy = 93% | [97] |
| | FT-NIR | Coconut oil adulteration identification | Band: 780 nm–5 μm | $R^2$ = 0.983 | [98] |
| | PLS-DA + VIS-NIR HSI | Nectarine variety identification | Band: 400–1 000 nm | Accuracy = 96.3% | [99] |

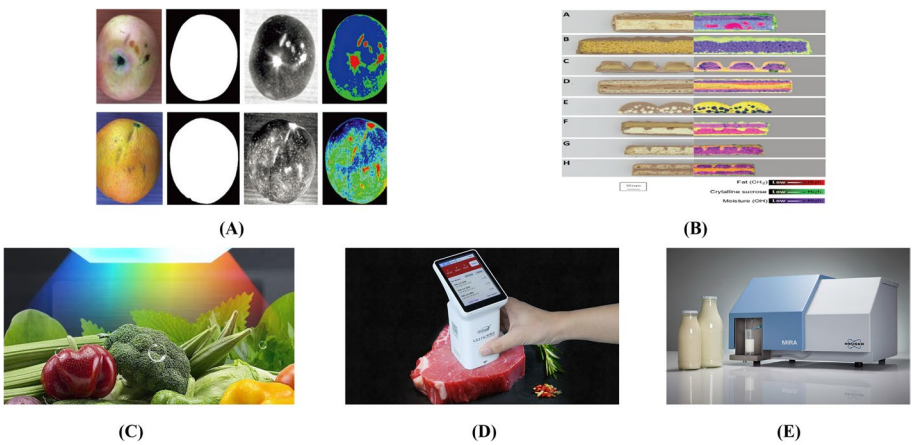



**Fig. 3** **A** External quality evaluation; **B** Internal quality evaluation; **C** Nutrient component analysis; **D** Freshness and safety detection; **E** Adulteration identification and authenticity analysis

SERS and fluorescence imaging technology, various spectroscopic technologies play their unique advantages in the quality detection of different agricultural products.

Studies on the quality detection of various agricultural products have shown that spectroscopic technology can be applied in multiple fields of agricultural product detection, such as external quality evaluation, internal quality evaluation, nutrient component analysis, freshness and safety detection, and adulteration identification. By using modeling methods such as LS-SVM, PLSR, and ANN, combined with appropriate spectral pretreatment and variable selection technologies, high-precision prediction models can be effectively established to meet the needs of agricultural product quality detection.

Although spectroscopic technology has achieved certain results and progress, it still faces challenges such as poor model universality, high equipment cost, and influence of environmental interference. In future research related to this field, a series of measures should be adopted to further enhance the application level and practical effectiveness of spectroscopic technology in the quality detection of agricultural products, such as developing miniaturized equipment, promoting multi-technology integration, introducing artificial intelligence algorithms, and establishing standardized databases. With the continuous progress and innovation of technology in the future, spectroscopic detection technology will play a more important role in the field of agricultural product quality detection, provide strong technical support for agricultural product quality and safety and industrial upgrading, and promote the rapid development of the agricultural product industry chain.

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**Authors' contributions**
Jiangao Qiu: Conceptualization, Writing—original draft, Writing—review & editing. Xirui Zhang: Conceptualization, Investigation, Supervision. Jie Guo: Conceptualization, Investigation , Writing- review & editing, Supervision. Tao He: Investigation, Data curation, Resources. Lina Zhang: Investigation, Supervision. Shaohua Fu: Resources, Supervision. Ronglei Pan: Resources, Datacuration. Keyan Jia: Resources, Writing- review & editing.

**Data availability**
No datasets were generated or analysed during the current study.

## Declarations

**Competing interests**
The authors declare no competing interests.

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